

Module 8: Working Capital Management

Topic 145-Working Capital and Its need in business

In simple words, **Working Capital** is that part of total capital which is used in the operations of a business. For example, the amount that is invested in a business initially is put into the Fixed Asset. It means that a larger part of our capital is invested in our fixed assets (I.e., land, building, equipment, machinery, vehicles etc.).

But once we have invested some of our capital in these fixed assets then we also required Additional Capital to generate profit from these fixed assets. And this Additional Capital is required for the purchase of inventory, for blocking our money in the accounts receivables that is debtors, whenever we make credit sales to make prepayments, to have sufficient amount of cash in our hand.

So, when we look at all these assets we collectively termed them as Current Assets. By definition, **Current Assets** are those assets which are normally kept in the business for a period less than one year. And it does not mean that if any of the current assets is kept in a business longer than one year, it will not be a current asset but by definition we call those assets which generally have a life of one year as Current Assets.

And likewise, opposite to current assets we have got some liabilities, these are those liabilities which are to be paid within one year time period so, we call them **Current Liabilities**. So, Working Capital in its nature is basically about Current Assets and Current Liabilities.

Gross WC (Working Capital):

Gross Working Capital is one in which we only consider the Current Assets meaning that if we add our all Current assets then we can name that figure as Gross Working Capital.

Net WC (Working Capital):

These items of Current Assets are to be actually financed mostly by Current Liability. So, if we take a difference of our Current Assets and Current Liabilities then it is called Net Working Capital.

Net Working Capital = Current Assets – Current Liabilities

Requirement or Importance of Working Capital:

So, this is the definition of Working Capital and simply its importance that if as a business organization we do not have enough Working Capital then we will not be able to operate and run our business.

It means that our Fixed Assets (plant, machinery, vehicles, equipment, land, building etc.) are useless even till if we do not have inventory of Raw material through which we produce finished goods and sale in the market or if we do not have enough money for the payment of electricity bills to run these machinery. So, this is the requirement of Working Capital.

Why we need Working Capital?

As we have said that Working Capital comprises of Current Assets so let's have a look at what are the major main Current assets which are mainly there in almost every business organization.

Why we need Working Capital?

- Inventory
- Receivables
- Prepayments
- Cash



Inventory:

We need Working Capital to operate the business because without Working Capital, our machines will not run; they will not produce anything which can be sold. So, we need to have inventory means we should have stock in trade. In manufacturing business where we convert raw material into finished goods so, we have got Inventory of Raw Material. And that Inventory is required in the business; if we are doing a trading business that we call Merchandizing business then there Inventory is that Finished product which we purchase and sell them.

But in any case in our business, which is not a service-oriented business we need Financing and Working Capital for Inventory.

Accounts Receivables:

We also need Working Capital for Accounts Receivables.

Why we need Working Capital for Accounts Receivables?

When we do credit sales or borrowing then our product on which our cost has incurred and then sell it to someone without collecting or receiving cash from our customer. So, our capital or money or financing will be blocked with the customer till we don't receive our cash or payment of credit sales.

We need Working Capital to block that amount and which is obviously necessary because without credit sales we cannot generate income and hence we cannot generate profit.

Prepayments:

We also have to make several Payments in advance and if we do not make these payments then we cannot operate our business. There are some mandatory Prepayments so, we need to have Capital and which is the Working Capital that is to be used for Prepayments.

Cash:

We also need Working Capital to have a sufficient amount of Cash in hand.

Why we do need Cash?

Cash is something which is considered as the most liquid asset. Here the most Liquid Asset means that such an asset which can be used in any work or any kind of payment or discharge of liability or incurring of expenses or purchase of any goods or assets. So, the asset which is more liquid is Cash and as Cash is required in everyday operations (I.e., for the payment of electricity bills, salaries, traveling, entertainment, repair & maintenance, purchases of goods etc.).

If we do not have sufficient cash then that may hinder the operations of the business and ultimately this may result into a negative effect on our business.

So, we can conclude our discussion on **Working Capital** that we can say that Working Capital is the essential portion of capital which is required to run and manage a business. Without Working Capital, there is no concept of having a successful operating business. It includes; current assets which are mainly the Inventory, Receivables or Debtors, Prepayments and also Cash.

A part of that Working Capital that is Current Assets is finance to Current Liabilities and in order to calculate the Net amount of Working Capital, we actually deduct the Current Liabilities from the Current Assets and we call it **Net Working Capital.**

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Topic-146-Approaches of Working Capital management

As we have discussed earlier that **Working Capital** is that amount or portion of capital which is used to operate or run a business. **Working Capital** is like blood in the body that is; if blood is not circulating in the body then we can say that the body is dead.

So, if in a business we do not have Working Capital which actually circulates in a business so, we can say that this business is dead. To continue a business or operate a business and then later on to make it grow further we need to have Working Capital.

But the question is that how much Working Capital should a business organization have? Is it good or bad to have more Working Capital? And what are the positive and negative points of having Working Capital more than our requirement or less than our requirement?

That is how we can manage our Working Capital.

In this lecture, we will learn about the basics of three major approaches of Working Capital Management.

Major approaches of Working Capital Management:

Hedging Approach:

Hedge is a protection which is generally there around something or some area so, this term **Hedging** actually is used as protection means protecting something against something.

So, when we talk about **Hedging Approach of Working Capital Management**, then here we say that we are trying to meet some requirements with some specific types of financing or we are hedging a certain requirements with similar kind of financing instrument.

Here two things to consider is that; Requirement or the nature of Requirement and the second thing is the **Type of Financing**. There is some such part of Working Capital that we can call **Permanent Requirement of Working Capital** meaning that throughout the year or operations of the business, we always need that amount of Working Capital in our business. We cannot go down from that amount and if we go down then our business will adversely be affected.

And we have got certain Requirement which is called a **Temporary Requirement of Working Capital** and in other words we can say it Seasonal Requirements meaning that these requirements are available sometime and sometimes we do not require this Working Capital.

We have to have financing to fulfill the requirement of this Working Capital. Financing can be through Equity or through Debt. Equity is always long-term

financing because Equity is not generated for some days or months but Debt can be Long-term or Short-term.

So, generally we meet the requirement of Working Capital through the Debt. If the Debt is long-term and on the other side we have got some Permanent requirement of Working Capital and then we are matching it with the long term financing. It means that if we have our Permanent or long term Requirement of Working Capital and same as it we generate long term financing and as our Temporary or short term Requirement and same as it we generate short term financing then such kind of working capital approach is called Hedging Approach.

Conservative Approach:

Conservative is in the sense that we try to have enough amount of financing so that, we do not run short of financing. It means that in this approach, our effort is that our financing will never be less than our requirement and if it becomes more then its acceptable. So, we make a blend of Long term and Short term financing for Permanent and Temporary requirements of Working Capital in a way that we increase the proportion of long term financing and reduce the proportion of short term financing.

It means that in Conservative approach, we have more long term financing than short term financing.

Aggressive Approach:

Here the term Aggression is in terms of maximizing our profit or in terms of utilizing our resources at maximum level. So, in this approach of Working Capital Management which is the third approach; the mix of short term and long term financing in such a way that relatively we have got lesser long term financing and we have got more short term financing.

When we conclude our discussion about the three approaches of Working Capital Management then;

In Hedging Approach, we arrange our short term and long term financing according to our Permanent and Temporary Requirements of Working Capital.

In case of **Conservative approach**, we generate more long term financing and generate less short term financing.

And in **Aggressive approach**, we generate or depend more on short term financing and reduce long term financing.

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Topic 147-Hedging Approach of Working Capital Management

In the previous lecture we have talked about **Working Capital Management's approaches** and discussed that there are three major approaches of Working Capital Management namely the **Hedging Approach**, the **Conservative Approach** and the **Aggressive Approach**.

And in that lectures we have talked about the requirement or the nature of the requirement of Working Capital and we said that some part or portion of our Working Capital is always needed in the business. There is no time where we do not require it.

But in contrast to this, we also have some portion of Working Capital which is not always needed or which is a Seasonal Requirement or Temporary Requirement. On the other side, we have got the financing options with us and these financing options are long term and short term.

Now the important thing is that **if we go for short term or long term financing then how would it affect our business' profitability or the cost of the business?**

Long term financing is most costly meaning that rate of interest or rate of return on the long term financing that a business has to pay, is more as compare to Short Term financing. It means that if a business increases the proportion of long term financing then overall cost of financing will also get increase.

Hedging Approach:

We said that Hedging Approach is one which matches the requirement of Working Capital with the type of financing.

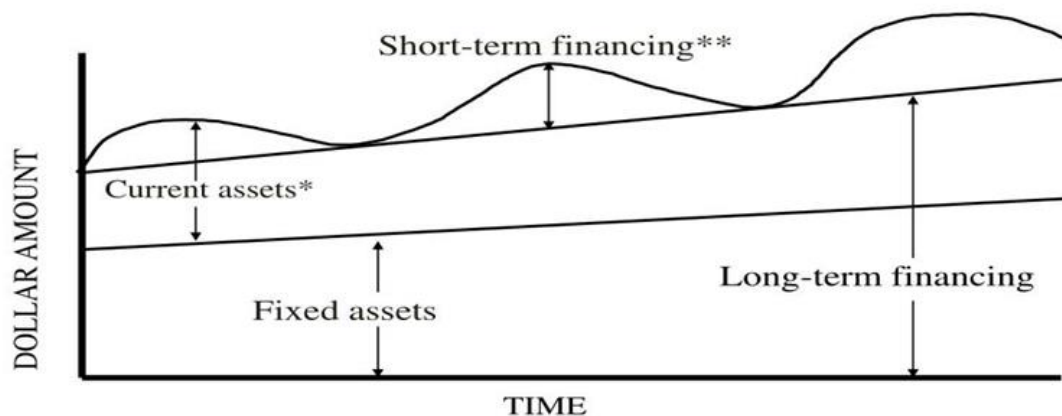
Short term financing for Temporary Requirement: So, short term financing is generated for Temporary Requirement of Working Capital. It means that such Seasonal Requirement which is sometimes available or sometimes not available so that the time period for which Working Capital is required and then arrange short term financing for that time period and when this requirement has been finished then we pay-off that short term financing.

Long term financing for Permanent Working Capital Requirement: And that portion of Working Capital which is always needed or Permanent Requirement of the business, we go for Long term Financing. As we require that Working Capital whole year so such financing is generated for that requirement which should be available with us.

Let's have a look at the graph that will explain that how working capital approach of Hedging actually fulfills the requirements of Working Capital with similar kind of financing.

Hedging or Maturity Matching Approach

A method of financing where each asset would be offset with a financing instrument of the same approximate maturity.



So, in Hedging approach, we go for short term financing only for that Temporary Requirement of Working Capital.

Here in this graph, long term financing is not only fulfilling the whole requirement of Fixed assets but also arranged for that portion of current assets or working capital which is our **Permanent requirement**.

So, in this graph, we can see that how Hedging Approach of Working Capital Management works. Here we can say that as our requirement which is long term or permanent and similar is our long term financing for that requirement. And as our requirement which is short term or Temporary then similar will be our short term financing for that requirement. And when this requirement has been finished then our short term is not outstanding.

What is the advantage or disadvantage of this?

Its advantage or disadvantage is **matching** that we have such financing which we is our requirement and that will not be more or less than it. It means that for days for which short term financing is required and that short term financing will be available for these days and after that this financing will not be available.

So, we can conclude our discussion is a way that a mix of Short term financing or long term financing is made in this **Hedging approach** that in which we match our financing with our Permanent or Temporary requirements and there create an overall cost of financing through this matching meaning that in this approach we will have a mix of the cost of short term or long term financing that we can calculate by taking the weighted average and we can see that how much will be the overall cost of the financing of Working Capital under the Hedging Approach of Working Capital Management.

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Topic 148-Conservative Approach of Working Capital Management

We have discussed in previous lecture about Working Capital Management's one approach (Hedging approach) in detail and we have seen that what the Working Capital Management is. We have learnt that Working Capital is such an essential requirement without which any business cannot be run.

We have two things to manage this Working Capital that we have to consider; one which is the Nature of Requirement (e.g., Temporary or Permanent) and other is Types of Financing (e.g., Short term or Long term).

We have seen in Hedging approach of Working Capital Management that in any business if we will require permanent or long term requirement of working capital then we arrange same long term financing for it. And similarly, if we will require temporary or short term requirement of working capital then we arrange same short term financing for it. It means that we match the requirement with the similar type of financing instrument.

Now we will see that what Conservative Approach of Working Capital Management is.

Conservative Approach:

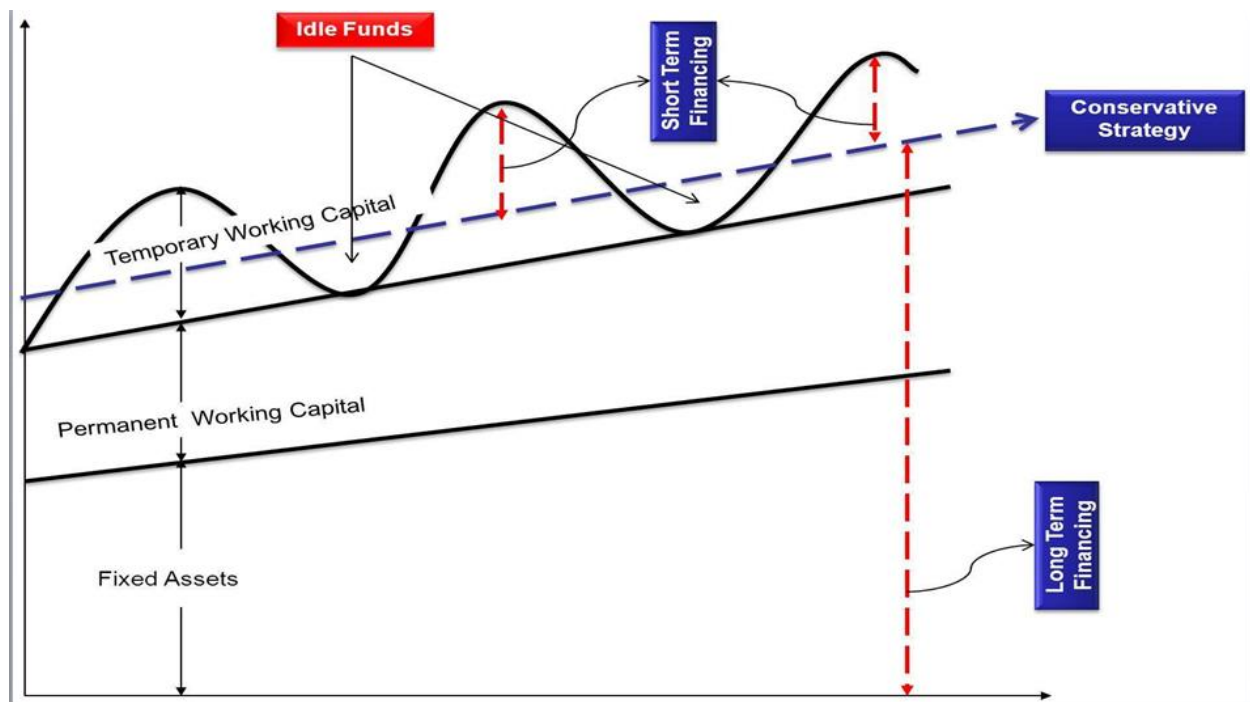
Some part of temporary requirement of working capital is met by long term financing

As we have said that Conservative Approach means that we would like to have surplus cash or surplus working capital always available with us in order to have a view point that in any situation we face shortage.

So, to avoid any possibility of shortage of working capital, we act little Conservative therefore we keep some working capital with us even when we do not need that working capital. For this, we meet the some part of temporary or short term or seasonal requirement of our working capital with the long term or

permanent requirement. In Hedging approach, we only take long term financing for permanent requirement but in Conservative approach we take some portion or some part of our temporary requirement is also met through long term financing.

We can understand that when we will have a mix of short and long term financing and the proportion of long term financing will be more, our cost of financing will be higher.



So, we can say that in Conservative approach of Working Capital Management, our profitability becomes less as compare the Hedging approach because our short term financing is less and long term financing is more and that more long term financing increase our cost of financing.

In conservative approach the amount of profit is minimum and the cost of financing is maximum.

Why any financial institution charge more return or interest from us for long term financing?

Its basic reason is that whenever financing is given for longer term, the uncertainty or the risk element increases meaning that if we are given anyone a loan for 3 months so we can forecast the possible changes in the next three months and we can be pretty sure or we can anticipate that where we would be standing 3 months from now.

But if we are providing financing for longer time period like for 5 years or 10 years or more, then a lot will change in that period i.e., economic, political, social and also business conditions so there will a lot of uncertainty when we talk about long term financing.

That is why, to cover that additional element of risk, banks and financial institutions would charge more interest or they would like to have more return on their financing. So, long term financing becomes more expensive than short term financing.

So, we can conclude our discussion of **Conservative approach** of working capital management that we can say that in Conservative approach, the amount of profit that we are generating from that business will be minimum due to less short term financing and more long term financing.

And our cost of financing will become increased and we know that cost and profit are inversely proportional to each other as when cost of financing increases then automatically profit decreases.

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Topic 149-Aggressive Approach of Working Capital Management

In our previous lectures, we have learnt about two approaches (Hedging approach and Conservative approach) of Working Capital Management. We have seen in these two approaches that how long term and short term financing has been arranged for permanent and temporary requirement of working capital.

So, Working Capital Management approaches are all about the Nature or type of financing and Nature of Requirement of working capital.

If we have temporary and permanent requirements then on other side we have short term and long term financing. In addition to this, long term financing is more expensive than short term financing meaning that if any business keeps on increasing the proportion of its long term financing then its overall cost of financing will also keep on increasing. And hence this increase in cost would reduce the profit or that business.

It means that adopting a certain approach of working capital management is not just a matter of managing working capital but it has a tangible effect on our profit meaning that if we adopt one approach then our profit becomes decreased and in other approach our profit becomes increased.

Now we have to discuss about the third approach of Working Capital Management that is Aggressive Approach.

Aggressive Approach:

Some part of permanent requirement of working capital is met by short-term financing

As we have said that in Conservative approach, if the proportion of long term financing has been increases so that we have got a capital available with us that we have always got some financing even if we don't need it. It means that if our calculation of seasonal requirement is out or if our seasonal requirement has been increased for some time then we must have a capital through which we meet our

financing requirement. So, that is **Conservative approach** where more is the long term financing and less is the short term financing.

In **Aggressive Approach**, we try to maximize the profit meaning that here objective is that how to minimize cost of financing and maximize the profit. For that, we need to do is that we need to alter the mix of long term financing and short term financing. We need to reduce the portion of long term financing in order to reduce the cost of financing and increase the proportion of short term financing to reduce the cost.

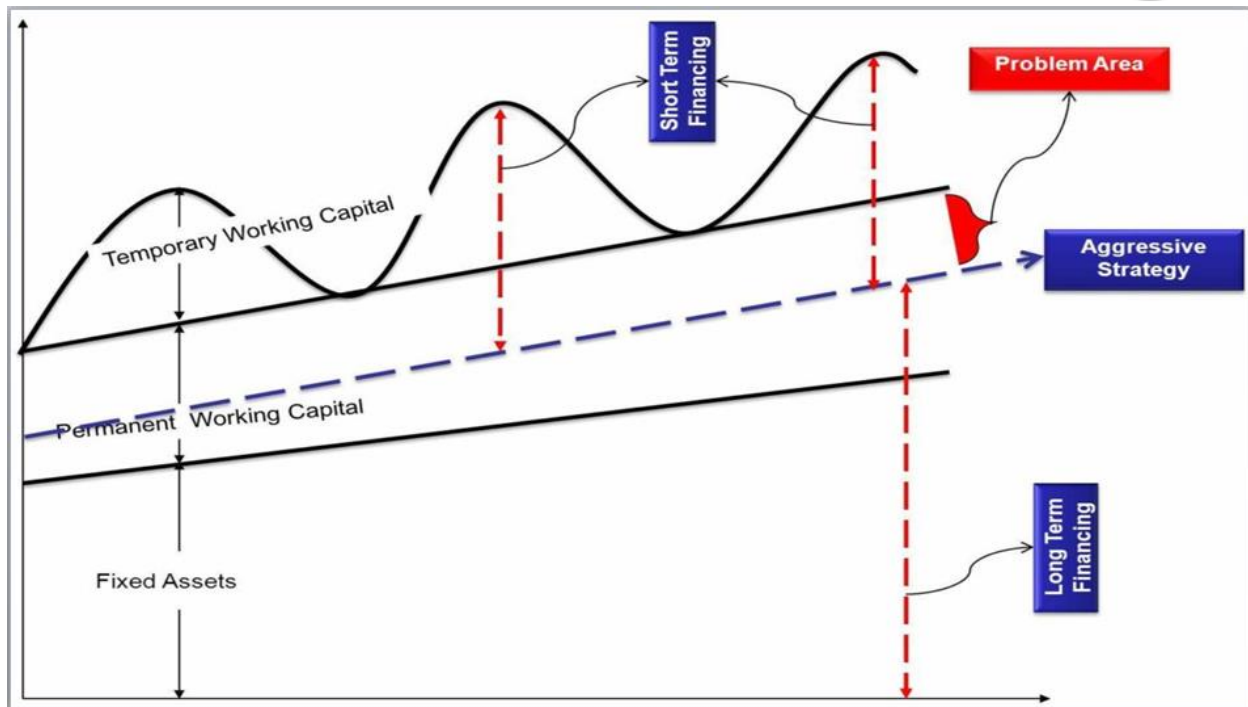
By doing so, we are meeting some part of our permanent requirement of working capital with short term financing. It will be a difficult task as all the time we need to have short term financing available for permanent or long term requirements.

The specialty of short term financing is that its maturity is for some months meaning that if financing is taken for 3 months then after 3 months this financing agreement will be finished and in advance we have to have another arrangement which will start immediately after 3 months, so that we have no shortage of financing.

It means that when we are using short term financing for permanent or long term requirement then every time short term financing should be there with us otherwise consequences will be shortage of working capital. And due to the shortage of working capital results in slow down of production, curtail our sales, delay our employees payments and we can have all adverse effects on the business.

So, **in aggressive approach**, we do increase or try to increase the profit of the firm but we need to be as the management of that business organization has to be very vigilant and we should always have some financing arrangements in short term available for our permanent or long term requirements.

So, we can define this **aggressive approach** of working capital management as some part of the permanent requirement of working capital management is met by short term financing.



In aggressive approach profit is maximum as the cost of financing is minimum

Aggressive approach of working capital management is actually adopted to increase the profitability and though which we can enhance the profit of our business. This enhancement is actually be achieved by reducing our cost of financing. So, when we use more component of short term financing, we actually reduce the cost of total financing which ultimately results into enhancement of profit.

So, **Aggressive approach** is the best in terms of profit and **Conservative approach** is worst because it gives us the minimum profit and **Hedging approach** is a middle of the both approaches where we have got more profit than Conservative approach but less profit than Aggressive approach.

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Topic 150: Accounts receivable management-1 Credit policy

Today we are starting our discussion on Accounts receivable management. In the first part of these series of lectures, we will be talking about Credit Policy.

The term Credit refers to the selling our product or services without receiving cash. Whenever we allow credit to our customers or our clients there is a certain policy that we follow meaning that as a corporation or business, we have to have a certain policy regarding the credit that we will allow to our customer.

Credit is basically an incentive meaning that if we allow someone that he does not make payment at that time when purchase something but defer that payment and pay it in the future at some time period. So, generally this is an incentive to buy for a customer.

So, Credit period is certainly something which affects our sales. So, if allow credit to our customers, this is likely to increase our sales or revenue and hence ultimately it is likely to increase our profit.

Components or Essentials of Credit Policy:

Credit period and Collection period: When we do credit sales or we give our things to someone without collecting cash and that is the time when given a certain period of time for collection. This agreed upon time period after which a customer has to make payment to the company or the business, is called the Credit period.

At times we do have a certain agreed upon Credit period but the amount is not collected in that period and there are delays and hence in reality we have got another time period after which the amount of sale is collected, so we call it Collection period.

So, we can say that **Credit period** is a theoretical agreement and **Collection period** is an actual time after which we receive our money or financing.

Credit Terms-2/10, n/30: Another important thing that is the Credit Terms. In Credit Terms we include; Net period of payment. **2/10, n/30** means that net period of payment is 30 days, the customer must pay within 30 days but if the customer will make a payment within 10 days then he/she will get 2% discount. And this is called the **Credit Terms**.

Cash Discount: In these Credit terms **2/10, n/30**, this 2% discount which we offer to our customers for earlier payment, earlier than the net period of payment that is required inducing them to pay us earlier than that time period, this discount is actually called the **Cash Discount**.

So, in the terms **2/10, n/30** this 2% discount within 10 days will be the Cash Discount.

Maximum Amount of Credit: Then the maximum amount of credit that can be given to a customer is a part of the Credit Policy meaning that how much total you can do credit sales to a customer. Then we need to look into it that what is the Capability (how much this business has been increased, or how much money this business has then it can return to us and what is the level of its business operations etc.) of a particular business to repay to us.

On the basis of this Capability, we can decide that how much Maximum Amount of Credit sales can be made to any customer.

Credibility: The Credibility is actually that how good is the Reputation of a person or a business in the market.

Credit Worthiness:

Another term which is used for credibility is **Credit Worthiness**. The Credit Worthiness includes both the elements that is; how capable a person or a business is to make the payments meaning that how much is the capability in any business to pay off its debts.

On the basis of some points (How is the Cashflows, how is its profitability, how are its different liquidity ratios etc.) we assess that what is the Credit Worthiness of any business organization. And according to this Credit Worthiness, we make our Credit Policy that how much amount credit of can be given to this person or a business.

How to measure the Credit Worthiness of every business?

Credit Rating: We can measure the Credit Worthiness of any business through Credit Rating. As the students are awarded with A, B, C or D grades for their academic performance and similar to this, business organizations on the basis of their potential to repay their debts, their profitability, their liquidity etc. are given some rating and this rating shows that capable a business is to pay off its debts.

Companies which do Credit Rating (Standard & Poor, Goody's, PACRA):

This rating process is done by several agencies or companies or firms which has got expertise in developing this kind of analyzing the companies' performance and the standing and then awarding a certain grade or a level or a rating to that particular companies like Standard & Poor, Goody's, PACRA. These are the names of some of the companies which do these Credit Ratings.

Investment Grade and Junk: As a result of Credit Rating, we have got some businesses which are considered to be Investment Grade. Here **Investment Grade** means that these are such businesses whose profitability is good, whose balance sheet is also good, there is growth also, there is no liquidity problem so, all the better or the good elements are existing there.

And against that, we have got some businesses or organizations whose securities are considered as Junk that means that there are those organizations whose credit worthiness is very low and there is more risk or risk to lose our money to business with them.

So, we can determine Credit Worthiness or Credit Rating of a company or a organization by using these Credit Rating agencies ratings.

How these are ratings are performed?

Different rating organizations use different letters or different grades as given below:



We can conclude our discussion that Credit Rating tells us about the Credit Worthiness and on the basis of Credit Worthiness, we see that how much credit can be given to which customer and what Credit Terms can be offered to that customer at the time of making our Credit Policy.

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Topic 151: Accounts receivable management-2

We will continue our discussion on Accounts Receivable Management or Credit Policy. We have seen in previous lecture that Credit Policy is very important element because every business has to make sales and increase sales and in order to increase sales almost every business is bound to give a certain credit and that credit is a norm of that business.

It means that if any business is operating in any specific market and dealing with a specific type of customers then in that market, a certain time period of credit has been given so that business cannot compete unless that credit is given to the customer.

So, for that a Credit Policy in detail has to be prepared which would require the knowledge of Credit Worthiness means that whether this business has a better standing or it is getting lesser rating and according we see that how much credit can be given. We have also talked about Cash Discount, Net Credit Period and Collection Period.

Now we will see some other terms to understand Credit Policy whose application in our credit policy will be needed.

- Cost:

Cost is divided into two categories; Fixed Cost and Variable Cost

Fixed Cost: It is that cost which does not change with the increase or decrease of our level of operations. It means that if our sales go up then fixed cost will not change and if our sales go down then still fixed cost will not change.

The cost which is independent of the operational level of a business is called **Fixed Cost**. For example, fixed rent, fixed salary etc.

Variable Cost: In contrast to Fixed Cost, there is Variable Cost. It is that cost which increases or decreases with the increase or decrease of our level of operations.

So, the cost which is dependent or it is proportional to our level of operations meaning level of production or level of sales. If we increase our sales then our variable cost will go up and if we decrease our sales then our variable cost will come down.

- **Contribution Margin:**

The term of **Contribution Margin** is used for the difference of Sales and Variable cost. It means that if we subtract our Variable Cost (excluding the fixed cost) from our Sales or Revenue then the amount or profit that we obtain is named as Contribution Margin.

Formula for Contribution Margin:

Unit Contribution Margin

$$\text{Unit Contribution Margin Formula} = \frac{\text{Sales Per Unit}}{\text{Total Variable Costs Per Unit}}$$



Contribution Margin Ratio: We can also calculate a ratio of this Contribution Margin meaning that if we divide this Contribution Margin with the Amount of Sales then we get the Contribution Margin Ratio.

Formula of Contribution Margin Ratio:

$$\text{Contribution Margin Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

Formula of Variable Cost Ratio:



$$\text{Variable Cost Ratio} = \frac{\text{Variable Cost}}{\text{Sales}}$$

Opportunity Cost:

When we draw back our investment from any one place and invest it in any other place and from which place we draw back our investment then the return we will be getting from that place, we have to break or forego that return. So the return that we are leaving or breaking in order to get another investment somewhere else that is called **Opportunity Cost**.

So, the return that we have to forego in order to invest our money somewhere else or we can say that **Opportunity Cost represents the potential benefits an investor misses out on when choosing one alternative over another.**

How an Opportunity Cost can be calculated?

Opportunity Cost is actually a percentage or is a percentage return meaning that it is that percentage return which we can get from any other place but as we have drawn our investment from there therefore we cannot get this percentage return.

So, this Opportunity Cost will come in our calculations in the form of percentage.

We can conclude our discussion here that **Opportunity Cost** is actually our benefit or return or profitability and its percentage and **Cost either Fixed Cost or Variable Cost** is actually such amount which will go from our pocket as a business. So, comparing the Cost and Benefit (Opportunity Cost) we will be making decisions about the Credit Policy.

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Topic 152: Accounts receivable management-3

We have started our discussion on Accounts Receivable Management which is basically a discussion on Credit Policy.

Today, in this lecture, we will see that what are the Components or Elements of Credit Policy which we have to consider while shifting from one Credit Policy to other Credit Policy.

What do we mean by changing the Credit Policy?

A Credit Policy will be considered change if we increase or decrease the credit period and if we allow more or less discount or if we increase the amount of credit that can be given to our customer. It means that the change of Credit Policy means that we have to change the terms of credit.

As we have talked about Credit in the beginning that Credit is an incentive to buy. If we have got credit then we have got some sales and if we increase this credit period then our sales are likely to go up. So, if we are changing our Credit Policy then generally it is about increasing the credit period and anticipating that how much will be the increase in the sales because of increasing the credit period.

In simple words, we can say that if we have got 30 days credit period for our customer and if we increase it to 60 days, we expect that our sales will also go up by a certain percentage. How much it will be in increase in sales and how much our cost will increase from that increase in sales and how much will be our profit we will get against that sales. This is something which is a discussion Credit Policy Decision.

Credit Policy Decisions:

Points to be considered:

There are certain points to be consider while shifting from a given credit policy to another credit policy.

Credit Policy Decisions

Points to be Considered

- Addition Sales
- Additional Receivables
- Additional Margin
- Addition Cost

Additional Sales: We have to consider what will be the Additional Sales. When we are making credit policy and that time we don't have actual results but we do have past data or have some basis on which we can decide or anticipate or estimate that how much will be the change or how much will be the increase in our sales. It means that it is a policy that is not in absolute or exact term but it is an estimate which is scientific and realistic estimate.

So, the first this is that we will see by increasing the credit period how much will be our Additional sales. And these Additional Sales can be in terms of additional number of units that will be sold or the amount of additional revenue in term of rupees or dollar amount.

Additional Receivables: When our sales will increase then our amount of accounts receivable will also increase due to that increase. If we do Rs.100 credit sales and our receivables will be of Rs.40 and if doing the credit sales for Rs.200 and our receivables will be of Rs.80. And for those Rs.80 accounts receivables, we would require additional financing.

It means that if we are increasing our credit sales and due to this increase our profit will also increase but with it our receivables will also increase. And when receivables will increase then due to this increase we have to generate further financing to cover that portion of working capital.

And there will be some cost of that financing which we have to bear.

Additional Margin: In addition to this, we have to see that how much Additional Margin we will get with the increase of our sales and how much increase we will get in contribution margin. So, we have to consider the increase in the contribution margin.

Additional Contribution Margin is actually our benefit or that profit which we match with our increase in cost and then see that is our new credit policy will be beneficial (in term of increase in profit) for us or is increase in profit is more or increase in cost is more from this policy. So, on the basis of which we will make our decision about credit policy.

Additional Cost: The last component that we need to consider is Additional Cost.

So, we will be talking about Additional Sales because of change in the credit policy, Additional Receivables, the cost on these Additional Receivables and Additional Profit which is in term of Additional Contribution Margin.

Formula of Accounts Receivables Turnover Ratio:

Accounts Receivable Turnover Ratio

$$\text{ART Ratio} = \frac{\text{Net Credit Sales} \quad \text{Average Accounts Receivable}}$$

Formula of Additional Receivables:

Additional Receivables
Additional Sales
Receivables Turnover Ratio

So, we will conclude our discussion here that we will use all these formulas in doing the calculation of our credit policy and to reach at a decision where we will decide that have we to shift from our credit policy to the new one credit policy or not and that would be more profitable for us or it will reduce or cut our profit.

Module 8: Working Capital Management

Topic 153: Accounts receivable management-4

We have discussed the Accounts Receivables Management and the Credit Policy and we have talked about in detail about what is the credit policy and its terms, Credit Period, fixed and variable cost, contribution margin, contribution margin ratio, variable cost ratio and opportunity cost in previous lectures.

Now we have to apply all these calculations in the calculations of credit policy that we have learnt in our previous lectures about credit policy.

So, we have got an example, which will help us to understand that how do we calculate the Additional Benefit and Additional Cost of changing from one credit policy to another credit policy. That is generally by extending the credit period that is; if we increase the credit period, we generate some additional sales and as a result we have more profit in the form of margin but we also have to incur more cost.

Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales = 25%	Rs. 600,000
Additional Units = 600,000/10	60,000 Units
Opportunity Cost	20% per annum

Contribution margin = sales – variable cost

Contribution margin = Rs.10 – Rs.8 = Rs.2

Additional sales = 25% of 2,400,000

$$= \frac{25}{100} \times 2,400,000$$

$$= 25 \times 24000$$

$$= \text{Rs. } 600,000$$

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs. 2 × 60,000 units = Rs. 120,000
Additional receivables = (Additional sales revenue)/(Receivable turnover for new customers)	Rs. 600,000/6 = Rs. 100,000
Investment in additional receivables = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs. 100,000 = Rs. 80,000
Required before-tax return on additional investment = (Opportunity cost) × (Investment in additional receivables)	0.20 × Rs. 80,000 = Rs. 16,000

Receivable turnover ratio = 6

As it is calculated as: $\frac{\text{total days in a year}}{\text{number of days allowed in the credit period}} = \frac{360}{60} = 6$

Percentage of variable cost = 80% = $\frac{80}{100} = 0.80$

Percentage of opportunity cost = 20% per annum = $\frac{20}{100} = 0.2$

So, we can conclude that if we change our credit policy, if we shift from 30 days credit period to 60 days credit period then our additional benefit will be a lot more than our additional cost. And hence our decision or our recommendation to the management will be that we should move with a new credit policy and change our credit period from 30 days to 60 days.

Module 8: Working Capital Management

Topic 154: Accounts Receivable Management-5

In the previous lectures, we have talked about Credit Policy. The main discussion of Accounts Receivables is actually about the discussion of Credit Policy. We have seen that Credit Policy has several elements related to credit period, percentage of discount and related to the additional benefits that a business would get by extending the credit policy.

In the last lecture, we have talked about the comparison of the benefit of changing a credit policy or rather extending the credit period in a way that we earn more return from that or earn more profit out of that.

We have talked about the concept of Contribution Margin and said that when we change credit policy or extend credit period then our sales also increases with this increase. And this increase in sales is actually results into more contribution margin which is our profit.

Then we compared this Additional contribution margin from the Additional sales that arises because of the change in the credit period with the opportunity cost of additional receivables which will be generated because of increase in sales.

Now, it is a next step towards the application of extension of credit policy. In the first case, we will assume that the new credit policy will only be applicable to the new customers. But if we think about it that as a company or as a business, we have increased the credit period and we have not allowing our existing customers to be benefited from that then it seems what workable.

It means that our existing customers will also wish to have more credit period as more credit period has given to new customers. In this regard, we would see that if a new credit policy is also to be applied along with the new customers off course on the existing customers then what will be the set of calculations for comparison of Additional benefit and Additional cost.

Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales = 25%	Rs. 600,000
Additional Units = 600,000/10	60,000 Units
Opportunity Cost	20% per annum

Contribution margin = sales – variable cost

Contribution margin = Rs.10 – Rs.8 = Rs.2

Additional sales = 25% of 2,400,000

$$= \frac{25}{100} \times 2,400,000$$

$$= 25 \times 24000$$

$$= \text{Rs. } 600,000$$

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs.2 × 36,000 units = Rs.72,000
Additional receivables of new sales = (New sales revenue)/(New receivable turnover for new customers)	Rs.360,000/6 = Rs.60,000
Investment in additional receivables of new sales = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs.60,000 = Rs.48,000
Level of receivables before credit period change = (Annual credit sales)/(Old receivables turnover)	Rs. 2,400,000/12 = Rs.200,000
New level of receivables associated with original sales = (Annual credit sales)/(New receivables turnover)	Rs. 2,400,000/6 = Rs. 400,000
Investment in additional receivable associated with original sales	Rs.400,000 - Rs.200,000 = Rs.200,000
Total investment in additional receivables associated with original sales	Rs.48,000 + Rs.200,000 = Rs.248,000
Required before-tax return on additional investment = (Opportunity cost) × (Total investment in additional receivables)	0.20 × Rs.248,000 = Rs.49,600

New Receivable turnover ratio = 6

As it is calculated as: $\frac{\text{total days in a year}}{\text{number of days allowed in the credit peirod}} = \frac{360}{60} = 6$

Old Receivable turnover ratio = 12

As it is calculated as: $\frac{\text{total days in a year}}{\text{number of days allowed in the credit peirod}} = \frac{360}{30} = 12$

Percentage of variable cost = $80\% = \frac{80}{100} = 0.80$

Percentage of opportunity cost = $20\% \text{ per annum} = \frac{20}{100} = 0.2$

Here we can see that our Additional Contribution Margin or additional benefit is Rs.72,000 and the amount of return or interest or opportunity cost that we have to pay is Rs.49,600. And we see that our additional contribution margin is more than our cost.

So, we can conclude that by adopting this new credit policy our Net Contribution Margin has been increased whenever this is the case that we are likely to get more profit than the increase in cost, we say that this new credit policy is recommended. It means that we will be successful to further increase our profit through the increase in our revenue by applying this credit policy.

In nutshell, we can say that in all these cases we would recommend the new credit policy.

Module 8: Working Capital Management

Topic 155: Accounts Receivable Management-6

We are discussing about Credit Policy. The Accounts Receivable Management is actually about the Credit Policy. We have discussed several aspects of Credit Policy about the Credit Period, Cash Discount, Discount Period and the benefit that we can get by extending the credit.

We said that Credit is an incentive as when any business or company extends its credit period then it creates an incentive for buyers and this incentive actually results into more profit or more contribution margin because of increase in sales.

As we get additional profit or contribution margin with the increase in sales then in addition to this, our cost of additional receivables also increases.

Bad Debts:

There is another element which increases with the increase in receivables which increases our cost or expenses that is **Bad Debts**. Bad Debts are generally calculated as a certain percentage of the amount of credit sales or the amount of average receivables.

It means that if credit sales will increase then Bad Debts will also get increase and if the amount of average receivables will increase (which will definitely increase due to the increase of credit sales) then our Bad debts will also get increase.

Meaning that if bad debts also get increased then in the calculation of our credit policy, we have to see that by increasing the credit period, if are getting more

margin then what is the increase in the amount of Bad Debts along with the cost of receivables.

Let's have a look at this example.

Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales = 25%	Rs. 600,000
Additional Units = 600,000/10	60,000 Units
Opportunity Cost	20% per annum
Bad Debts Expense	10%

Contribution margin = sales – variable cost

Contribution margin = Rs.10 – Rs.8 = Rs.2

Additional sales = 25% of 2,400,000

$$= \frac{25}{100} \times 2,400,000$$

$$= 25 \times 24000$$

$$= \text{Rs. } 600,000$$

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs. 2 × 60,000 units = Rs. 120,000
Additional receivables = (Additional sales revenue)/(Receivable turnover for new customers)	Rs. 600,000/6 = Rs. 100,000
Investment in additional receivables = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs. 100,000 = Rs. 80,000
Additional Bad Debts Expense = (Additional Sales × Bad debts percentage)	600,000 × 10% = Rs. 60,000
Required before-tax return on additional investment = (Opportunity cost) × (Investment in additional receivables)	0.20 × Rs. 80,000 = Rs. 16,000
Total additional cost = Bad debts + Opportunity cost	60,000 + 16,000 = Rs. 76,000

New Receivable turnover ratio = 6

As it is calculated as: $\frac{\text{total days in a year}}{\text{number of days allowed in the credit period}} = \frac{360}{60} = 6$

Percentage of variable cost = 80% = $\frac{80}{100} = 0.80$

Percentage of opportunity cost = 20% per annum = $\frac{20}{100} = 0.2$

Criterion for accepting or rejecting Credit Policy:

Here we have to compare Additional Benefit with the Additional Cost.

We can say that if Additional Benefit is more than Additional Cost then we will accept new Credit policy and if Additional Benefit is less than Additional Cost then we will reject new Credit policy.

When we compare the Additional Benefit that is Rs.120,000 with the Additional Cost that is Rs.76,000 then our Additional Benefit is lot more than our Additional

Cost so we would recommend that new Credit Policy should be adopted even when there is an increase in the cost of Bad Debts of Rs.60,000 and this is being the Criterion of accepting or rejecting Credit Policy.

Impact of Cash Discount in Credit Policy:

As we have learnt in previous lectures about different terms of Credit policy where we said that 2/10, n/30 actually shows that if we want to take 2% discount then we have to make payment within 10 days and if we will not make payment in 10 days then we have to make payment in 30 days which is the **Net Credit Period**. So, practically what happens is that there is a good percentage of our accounts receivables, our customers who would like to avail that discount.

It means that those people who want to make payment within 10 days and will get 2% discount. For those customers, accounts receivables who would pay within 10 days then the credit period will be 10 days and all their Receivable Turnover will be calculated according these 10 days that is $360/10 = 36$ and then this 36 will be considered only for that portion of accounts receivables who would avail a certain percentage of discount that we are offering.

It means that when we include the calculation of discount in credit policy then we have to anticipate that from our total Additional Sales how much percentage of people will be such which will avail discount. For these people, we will consider the time period on the percentage of discount and we will consider normal credit period for those people who will not avail discount.

And the calculations that we will do will be simultaneous meaning that on one side we do calculation of the people of discount period and on the other side we will do calculation of those people's calculations which have not availed discount.

As we have considered Opportunity cost as a cost and after that we have also included Bad Debts cost in it, similarly when a company allows a discount then discount also becomes a cost for the company.

So, **Discount** becomes third cost by combination with Opportunity cost and Bad Debt cost. And we will see our cost by adding these three costs and this third cost will be same compared with our Additional Contribution Margin or Additional

Benefit. And we can conclude by accepting or rejecting the Credit Policy if the Contribution Margin is more then we will go for the new Credit Policy and if the Contribution margin is less than that Additional cost then we would reject that Credit Policy.

Module 8: Working Capital Management

Topic 156: Inventory Management-1

Inventory or Stock is that asset which is used in a business for two purposes; one is that this inventory is of goods or merchandise meaning that such this that we purchase or sale and earn profit on them. So, one type of inventory or stock is **Stock in Trade** meaning that profit is earned by selling or purchasing that inventory or stock.

Types of Inventory or Stock:

Types of Inventory

1. Stock in Trade
 - a) Raw Material
 - b) Work in Process
 - c) Finished
2. Stores, Spares, and Tools



This stock can be of different types depending upon the nature of business. As we have learnt in previous lectures that there are some businesses which are called **Trade in Businesses, Merchandizing Businesses** where the job is to purchase some goods and sell them in the same firm without adding any value.

It means that as we purchase any goods or item of inventory then sell same as it is. And the difference in the selling and buying price is our **profit.**

So, in addition to that Trading or Merchandizing business, we also have got Manufacturing Concerns. **Manufacturing Concerns** are those purchase raw material or goods which are in the form of raw material for that production process and they apply labor and factory overheads and convert that raw material into finished goods. So, these finished goods are then sold in the market and profit is earned.

In between the process of conversion of raw material into finished goods, we also have got Inventory or stock of those items which are still incomplete and we call them **Work in Process.**

So, when we talk about **Types of Inventory** then we say that **there are two types:**

1) Stock in Trade:

In stock in trade, again for manufacturing concern we have stock of Raw material.

Raw Material: It may be that finished goods for any production process can be the raw material for any other production process. For example, textile spinning mills produces yarn or thread and the finished product of a textile spinning unit is yarn whereas if we talk about the textile weaving units, their raw material is that yarn and make clothes from that yarn. So, the finished product of weaving mill is grey cloth and the finished product of spinning mill is yarn.

So, **Raw Material is a matter of our production process that what are we going to produce.**

Work in Process: Then this raw material is converted into finished goods but in between it is in Process. When we are looking at the stocks on any given time, we will find some quantities of raw material, some quantities of those units which are incomplete units that are called **Work in Process.**

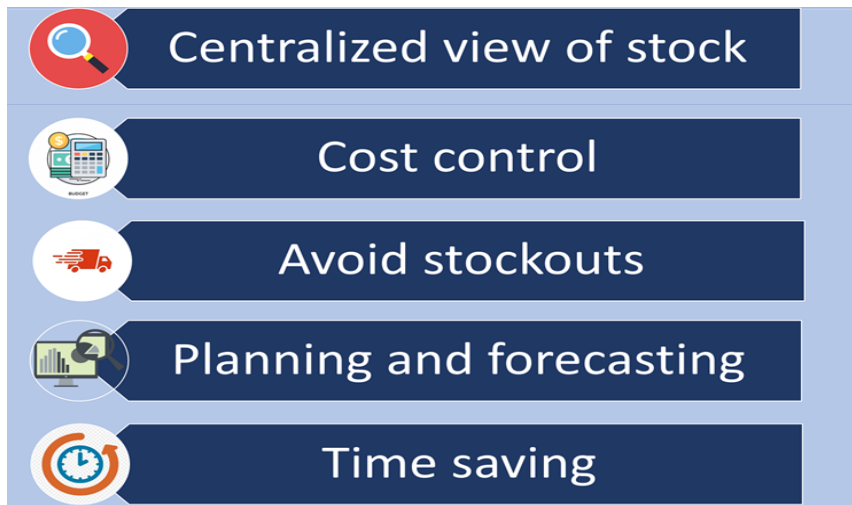
Finished: We also have some inventory of finished goods. Because as when we produce the finished goods then in there we do not sell them meaning that it takes some time to be sold and till their selling they kept with us.

2) Stores, Spares, and Tools:

The other type of stock in businesses or companies is called the **stock of Stores, Spares and Tools** meaning that this is stock of such things which do not sold but this is the stock or inventory which is kept in the business for internal purposes meaning that those things whose stock has kept for use in our business. This is also a **manufacturing concern**, a huge amount of inventory which is not for sale so, this is the inventory of Stores, Spares and tools.

Advantages of Inventory Management:

What is the purpose of Inventory management? Why would we like to have Inventory management and what is the benefit of it?



Centralized View of Stock:

Some of the points which are commonly there regarding the inventory management's benefit is having a Centralized View of Stock. When we talk about the larger business concerns, there are many inventory items which are there from which we use the bin cards, inventory cards, and lengthy process of purchasing those inventory items looking at various levels of the stock.

So, Inventory management is actually provides us the holistic view of the stock and inventory items.

Cost Control:

In addition to that, it provides us a Control because Inventory is something which also has a cost meaning that it is no free of cost to keep and maintain inventory but our cost incur on it which obviously reduces our profit. So, for better Inventory Management, we can have a better cost control and as a result we can enhance our profit.

Avoid Stockouts:

There is a situation which no company or no business would like to have where they have got orders, possible sales but they do not have inventory or products or goods available with them for sale and that **condition is called Stockouts**. So, we can avoid Stockouts by better inventory management.

Planning and Forecasting:

Inventory management helps us to plan and forecast the processes related to inventory or inventory items meaning that we can know in advance that how we will be there after some time? How much production will be possible for us? How many units we can sell?

Time saving:

Inventory management saves our time in terms of controlling, counting inventory, calculation of cost and ultimately this is something which also reduces our profit.

So, in Inventory management, we try to control the cost of stock, maximize our profit and ultimately this is something which results into the growth of any business organization.

So, we can conclude that Inventory Management is something which is likely to lead us towards better management of inventory and hence controlling our cost and ultimately increasing and maximizing our profit.

Module 8: Working Capital Management

Topic157: Inventory Management - 2

We have started our discussion about Inventory Management and have seen that Inventory Management is something which is essential for any business organization particularly for those businesses where they have got huge inventory with a many items in inventory.

So, the process of purchasing inventory which is called **Procurement Process**, recording the inventory and keeping the certain level of inventory with them so that we can avoid Stockouts, this is all part of Inventory management and it is extremely important.

It means that in any business organization, if they don't know about when they have to generate a process for purchasing the inventory items then they may face situation of **Stockouts**.

So, the **process of Procuring inventory** which starts with the Bin Card and **Bin Card** is a card which is kept on a cabinet or shelf where that inventory item has been stored. Whenever items of inventory are added in that Bin then Bin Card is updated and when the items are taken out of that Bin then Bin Card is also updated.

So, whenever this Bin Card actually shows a certain level which is called the **Level of Reordering**, the new order is generated meaning that a new Purchase

Requisition is generated and **Purchase Requisition** means that to raise a request in order to place an order to purchase goods.

And this **Purchase Requisition** has to pass through several hands and is to be approved by the people who are the part of the organization by looking at the quantity which is available, by looking at that production requirement and ultimately on the basis of this Purchase Requisition and order to purchase new inventory items is placed. That is called **Purchase order or PO.**

On the basis of it, the goods are delivered by the suppliers which are received in the store along with several documents which include the gate pass or the quality assurance report and ultimately these are placed in those Bins.

It means that this is such process through which we manage the inventory.

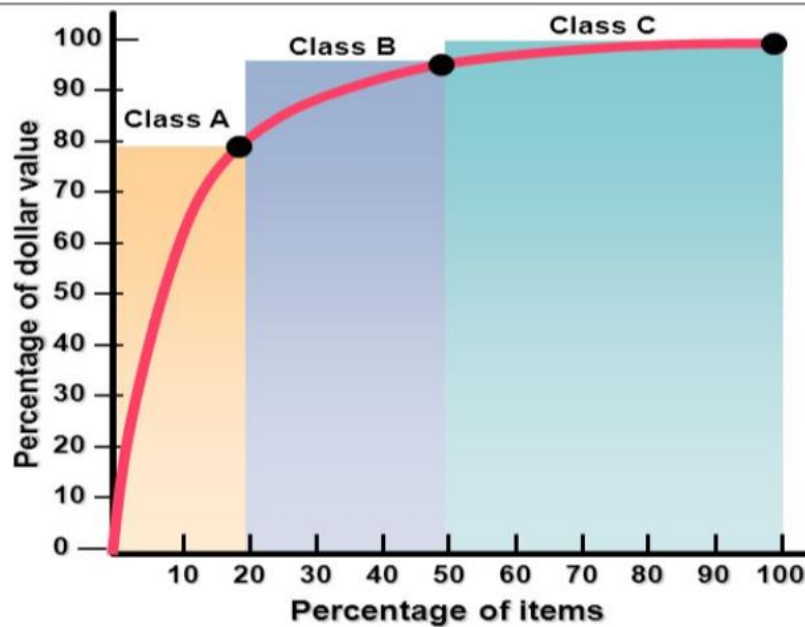
The **Basic Plan which is called the ABC Plan** is about managing the inventory with reference to the quantity of inventory and also with reference to the amount which is involved in holding that inventory.

We say that there are some inventory items which are very expensive but generally these inventory items are in smaller quantity. And then there are inventory items which are very cheap but these inventory items generally are in very large quantity.

But still the large quantity of those very cheap items would not require a lot of investment in inventory. And in between the two, there are some inventory items which are neither too expensive nor too cheap and they involve our investment.

So, we divide our inventory items into these three groups as group A, group B and group C where group A is very expensive inventory items whose number or quantity will be less but involve our more cost in them.

ABC Analysis



So, this classification is not just for the purpose of putting the inventory items into A, B and C category but this classification is also for the purpose that we need to take more care of those inventory items which are in category A meaning that if we purchase more items of category A than our need then our a lot of amount of working capital will be involved in it. And even if we purchase category C items more than our need so no more of our cost will not be involved in it.

So, any loss of category A inventory items is a huge loss. On the other hand, loss of C category items will not affect the company. So, the purpose of category A, B and C items is to group inventory items in three categories and then making policies and procedures accordingly so that we can control these inventory items in the best possible manner.

With reference to inventory management and stock levels, a very important part of our stock is the Safety Stock.

Safety stock is that stock which we always want to keep with us. This is for the purposes of securing ourselves of having a safety that in case if our new

shipment or new order does not reach to the production house in time then we should have this Safety stock with us.

We can conclude our discussion that if there is no Safety stock then we can go for Stockouts meaning that we face such situation that we don't have not stocks but customers or orders are present and bear loss.

In case of Stockouts, we have to purchase goods or inventory items then in that case we will have to pay higher price to buy those levels and hence our profits will be curtailed.

Module 8: Working Capital Management

Topic158: Inventory Management - 3

As we have learnt some of the basic things about **Inventory Management** and now we are continuing with some other things of Inventory Management.

Lead Time:

The term Lead Time is used for the time period between placing an order and actually receiving the goods. Because when an order is placed for purchasing goods then the goods are not purchased immediately and we have to wait for certain period of time. And this period of time is called **Lead Time**.

In some cases, this Lead Time is very short if the supplies are to be purchased locally and supplier is available easily. Then there may be Lead time of 2 hours or may be of day 1 or 2 but in other cases where the raw material or the goods to be imported from foreign countries then the Lead Time may be of few months.

Lead Time Usage:

During this Lead Time, we have to continue our production so, we have to consider another very important thing with reference to Lead Time that is the Lead Time Usage.

Lead Time Usage means that if we will place an order today and we will receive our goods suppose after 30 days so, we should have enough stock to continue our regular production for these 30 days. And that is how we plan to place our order.

So, these two terms are very important and very relevant in Inventory Management that are **Lead Time** (the time between placing an order and receiving the goods) and the other one is **Lead Time Usage** (the consumption of those goods or units during that lead time).

Economic Order Quantity (EOQ):

There is another important aspect of Inventory Management that is called the Economic Order Quantity (EOQ). Because we want to earn maximum profit and this is the objective of every business organization so that we need to cut off our cost as whenever we cut off our cost then we have to make it minimum and this is also the objective of Inventory Management meaning that inventory has to be managed in such a way that our cost related to inventory becomes minimum.

And for that the quantity that we should purchase every time when we place an order should also be the most cost effective quantity and we call it the **Economic Order Quantity (EOQ)**.

So, **Economic Order Quantity (EOQ)** is that the quantity that a business should order to keep it to the minimum level of Cost.

We consider **two things in Economic Order Quantity (EOQ)** which are: Ordering Cost and Carrying Cost.

Ordering Cost: Whenever a new order is placed then there is a cost to place that order meaning that what is the cost of placing this order as we have a warehouse or store department and there is a lot of documentation in this warehouse, there are storekeepers which keep record that when they have to make new purchase and there are many approvals that they have to take and after these approvals, that order has to be placed.

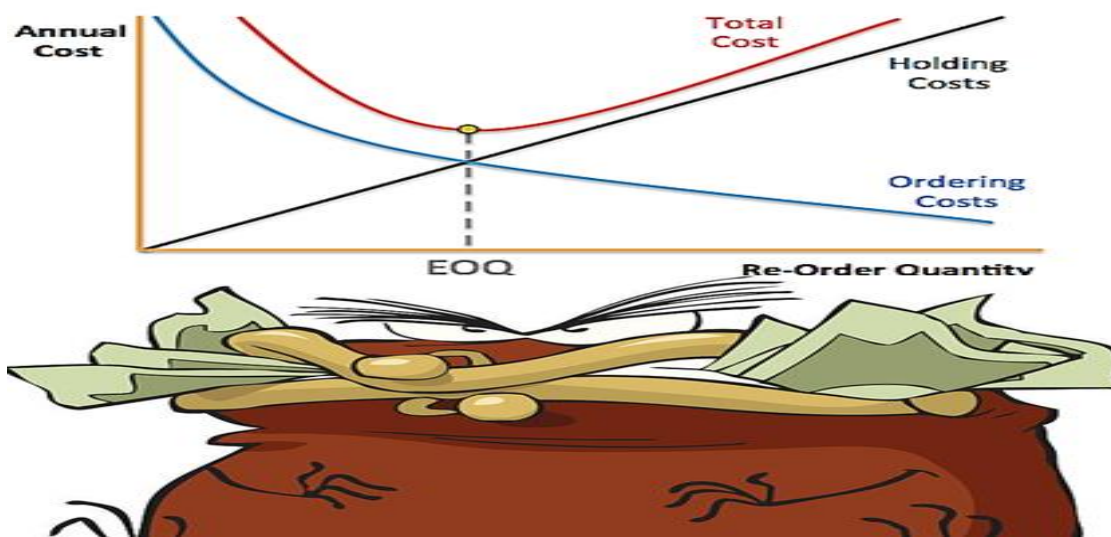
All this process actually requires a cost. So, every time when we place an order or new order then there is a cost that we have to incur which is called the **Ordering Cost**. It is obvious that the more is the number of orders that we are placing then more will be our Ordering Cost.

So, we need to keep those orders to the level that our total cost becomes minimum.

Carrying Cost: We can reduce the Ordering Cost by just placing one order in the whole year and purchase a lot of goods through which our Ordering Cost can be minimum. But on the other hand, we have a cost to carry those goods meaning that we also have to incur cost to hold or store or carry these units with us i.e., this is the cost of warehousing, we have to pay rent, we have to pay the overheads of that warehouse, we have to pay for security, cost for the people who are working there. So, that is also the cost which will be higher if the number of orders are lesser. If we increase orders then every time the quantity which we receive will become less and so in this way our Carrying Cost will starts to decrease.

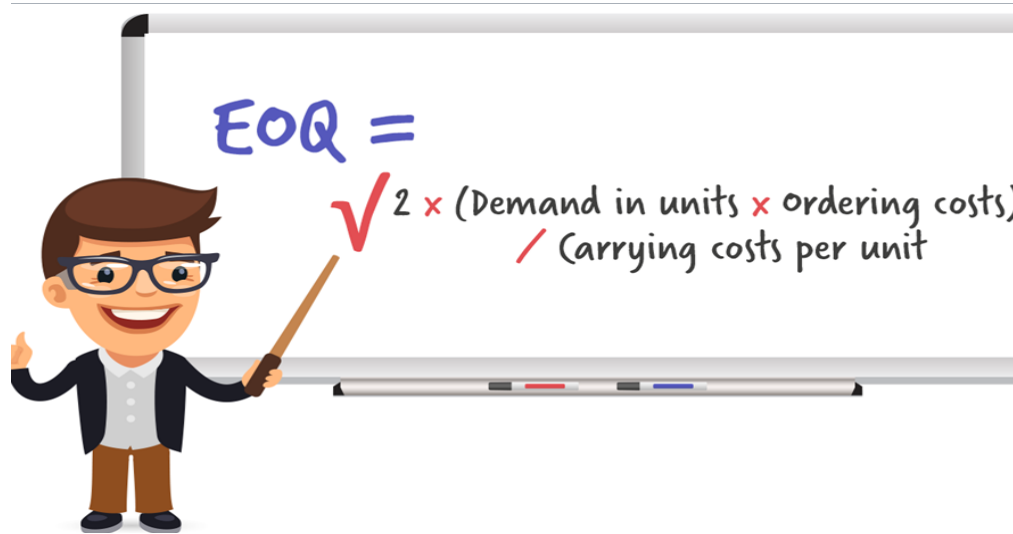
Total Cost: So, these two costs Ordering Cost and Carrying Cost are actually inversely proportional to each other and the sum of these two costs is the Total Cost related to Inventory.

Economic Order Quantity (EOQ) is that quantity of purchase for which the sum of our Ordering Cost and the Carrying Cost is minimum.



Here the graph of Total Cost shows that there is a point where some of Ordering and Carrying Cost is minimum even before that lesser quantity or higher quantity than that quantity will be more costly and this is the point which is called the Economic Order Quantity (EOQ).

Formula of Economic Order Quantity (EOQ):



We have talked about Lead Time and Lead Time Usage and we have talked about Economic Order Quantity (EOQ). To keep the cost to minimum level, we have to calculate a quantity at which our total cost will be minimum.

We can conclude our discussion that it is very important for every business organization to keep its cost to be minimum and for that they need to purchase every time a quantity for which their total quantity that is Ordering Cost plus Carrying Cost is minimum and that is called the Economic Order Quantity (EOQ).

Module 8: Working Capital Management

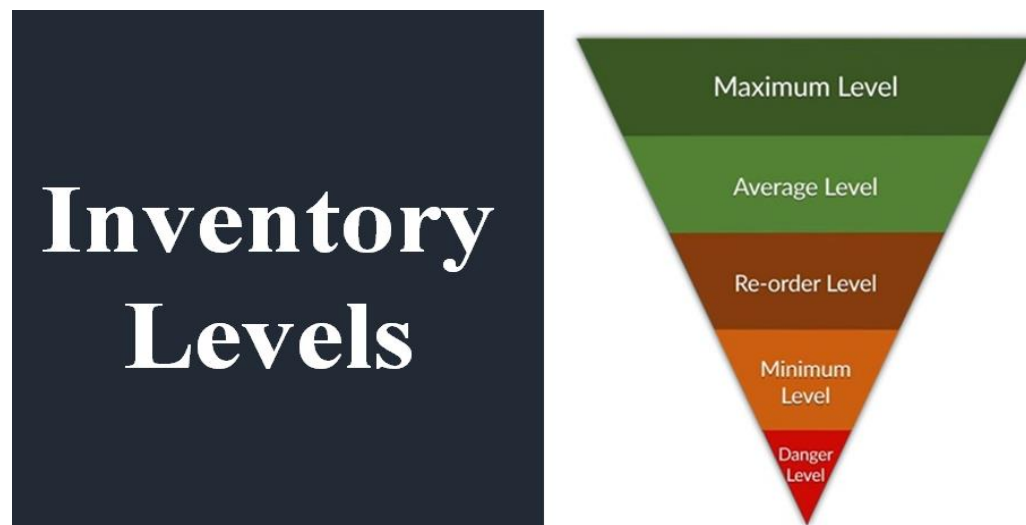
Topic159: Inventory Management - 4

In our previous lectures we have talked about various aspects of Inventory Management and we have seen that Inventory management is quite a complex process where we need to control the cost of inventory and we need to keep the supply of inventory in time.

It means that if we have shortage of supply then our production or sales may be affected due to which we can be in losses. If we purchase inventory more than our need or its quantity of purchase is not proper then it can also increase our cost and our profit can be down.

Levels of Inventory:

When we purchase inventory and keep with us then there are certain levels of these inventories.



Maximum Level of Inventory:

This is that level in which our inventory should not be more than that level and if we keep more inventory than this level then our cost inventory will become

increase and our inventory will be more than our need. There is a possibility that loss of inventory, damage of inventory etc. may start.

So, we set a level by calculating this level of inventory and we use a formula for this and we can determine from this formula that what is the maximum level of inventory that we should have?

Minimum Level of Inventory:

This is that level in which we should have inventory less than that level of inventory. We also need to set a level where the inventory should not be less than that level and that level is called **Minimum level of Inventory**.

How we can calculate these levels and how much will be the maximum and minimum level of inventory?

To calculate these two levels that are Maximum level and Minimum level by considering the requirement of inventory or by considering the production requirement or by considering the sales requirement.

So, from the production and sales requirement, we come up to the requirement of inventory and then we talk about that how much will be the maximum and minimum level.

Danger Level of Inventory:

When we say that there is a minimum level and there is a danger level then the question arises that what is the difference between these two levels.

Minimum level is the normal routine level which will be reached every time when we are purchasing inventory and using it in our production. It means that when we will purchase inventory and use it then our Minimum level will regularly be achieved and after that we will get inventory.

But at times it happens that the inventory or goods do not reach to the factory or to the business in times. As we have discussed about Lead time and Lead Time Usage that when after placing our order we are waiting for these goods or inventory that when they will reach and if they will be late then our inventory level will start to go down even from the minimum level. And it falls even below the minimum level

then we also have set another level of inventory which is called the **Danger Level of Inventory.**

We can say that **Danger level** is that level of inventory where we have to rush in emergency to purchase and to procure the goods/units so, that our production is not stopped.

So, we do not face the situation which is called Stockouts meaning that it is such a situation we have all things (labor, overhead, sales, demand etc.) for the production but we have not done production due to absent or goods or inventory.

This is the level when it is reached everything else is not more important than procuring that inventory and this level is called **Danger Level** and this is below the Minimum Level of Inventory.

Average Level of Inventory:

We can also calculate that how much inventory we do carry on average meaning that this Average Inventory Level will be the level which we should always have and in normal levels of production, this is the level of inventory that we maintain.

Reorder Level of Inventory:

Reorder level with reference to our Lead time and Lead time usage as we have discussed in our previous lectures. We determine our Lead time that how much days, weeks or months are there and how much will be our consumption of goods during this lead time.

We determine the Reorder level of our Inventory items by considering both Lead Time and Lead Time Usage which means that when the level has reached, we should place the new order and as a result we have enough units of inventory that we consume and as we touch Minimum level of inventory then we have new order and then again our level after inventory replenish and reach to our Maximum level.

So, **Reorder level** of inventory is determined by considering the Lead Time of the procurement of inventory and the lead time of usage.

We can conclude our discussion that there are various levels of inventory that one has to consider and calculate. Every level of inventory has its significance and these levels are maximum, minimum, average, danger and reorder level of inventory and all these levels are calculated and are kept there. So, the inventory could be properly managed, cost can be curtailed to minimum and production can be continued as it is planned.

Module 8: Working Capital Management

Topic 160: Cash Management-1

We are starting our discussion on the second element of Working Capital that is Cash.

When we talk about the term **Cash** then it means that the currency that we hold and the currency which is used in a country or internationally or the payment of expenses for buying or paying liabilities etc. In Pakistan, we have Pakistani rupees and internationally we have got dollars, yens, pounds and other currencies as well.

Every business organization has to hold and carry a certain amount of Cash and the purpose of holding this cash is to make the payments and in every business, every day we have to make several payments.

When we purchase goods, at times we have to purchase upon of those goods on Cash so, we require some amount of Cash with us. We have to incur several expenses which are to be paid i.e., utility expenses although we get some credit for that but after the credit period we need to make the payment, we need to pay the salaries, we need to pay to our staff, to pay loans, have to pay liabilities, to pay expenses of traveling and conveyance, entertainment, repair and maintenance etc. For all these things we need to have Cash. So, Cash is extremely important in a business.

But another thing which is important with reference to Cash is that holding Cash more than our requirement is not beneficial for a business from the view point of profitability meaning that if we save a lot of cash in our locker and it does not be

circulated in our business then as a result this cash will be idle or unproductive or we cannot earn further profit from that cash.

We can say that there are pros and cons that we need to have certain amount of cash but we should not have more than a certain amount of cash. Because then it becomes harmful for the business organization.

So, Management of Cash becomes extremely important as we need to determine that what amount of cash we should have at any given point of time and that much amount should be there neither more nor less than that.

When we use the term Cash then what do we mean by that?

- Cash in Hand
- Cash at Bank
- Cash Convertibles

Cash

When we use the term 'Cash' then his includes:

Cash in Hand: The Cash in Hand means that this is that amount of cash which is with the business organization in their premises meaning that this is a cash that we keep for performing our daily activities.

Another term which is used for this small amount of Cash that we carry every time with us is called the **Petty Cash** (The cash required for small expenses that we need to incur every day).

Cash at Bank: But bigger chunk of cash is actually there in the Bank because it is safe, secure and in times we also get some interest on the amount that we have for our bank account.

So, relatively larger amount of our cash holding is with the bank. But again as we have just discussed that the Cash has to be circulated so, the functional business will also not have a very large amount of cash placed in the bank.

Cash Convertibles: Another thing which is important we need to keep in our mind is called Cash Convertibles. Some of the items which are actually not the currency notes like not rupees or dollars but they are almost like Cash meaning that they are too much liquid as they can easily and quickly be converted into cash and can use them in payments or other expenses.

These items are called Cash Convertibles i.e., in Pakistan, we have got prize bonds so, some people would accept prize bonds as cash. There are some cards available which are called the **Plastic money** that is also included in cash.

Why we need Cash?

It is not possible to run a business without cash as we cannot perform regular operations of our business with having cash with us i.e., as we need to have cash for payment of expense, for purchases, for the payment of creditors, for the repayment of loans etc. So, for all these things it is essential to have cash with us.

If this cash will becomes less then what will we do?

Then we have to borrow cash and if we have to borrow cash then on that amount we have to payment some amount of interest and this interest is an expense which increases the cost.

So, **why we need cash** is the question that we should answering in our business from the view point of the requirements of cash meaning that where we need cash for performing our business operations.

Suppose we do not have enough cash and we cannot even borrow meaning that there is a situation arises that we do not have borrowing arrangement or banks are not ready to give us loan or our credibility/credit ratings is not such that we can do emergency borrowing so, sometimes it happens that a business which is apparently profitable which is making profits, goes into bankruptcy.

It means that it has to close its operations due to the absence of cash and this is the worst situation which a business organization would face meaning that no anyone

business organization want to have capacity of production, have sales demand but due to absence of cash they have to curtail their operations.

So, Technical Insolvency is a situation where having enough capacity to produce and sell and to make profit, still a business has to discontinue its operations and it is the worst possible scenario that we will have.

We can conclude our discussion that Cash is extremely important that without Cash, it is not possible for a business to operate. So, if it is not possible then the next question is that **what is the amount of cash that should be kept with us.**

So, to manage it, need to be calculated that what amount of cash should be with us in which month or day or week. Therefore we need the requirements of payments that we will make. Because we are into running a business where we earn some revenue, we have got sales, where we sell some of our assets where we will borrow also. So, all these things are actually the inflows of cash.

So, cash management is a process where we will see that from where we will be receiving cash and where we will have to pay cash and what are the appropriate levels of cash that we should have in order to efficiently and effectively run our business.

Module 8: Working Capital Management

Topic 161: Cash Management-2

We will continue our discussion of Cash Management and will see that what are those aspects which come under the discussion in Cash Management.

Why do we need cash?

We have done some discussion about it in our previous lecture as well and seen that where and why we need cash. And here we will be talking three basic motives of holding cash.

Motives of Holding Cash

- Transactional
- Speculative
- Precautionary



Transactional Motive:

Transaction is an activity of a business which can be shown in a monetary value. When we purchase something, when we borrow cash, when we deposit cash then, when we sell goods so, all these are transactions.

So, these transactions involve cash and the others do not. For example, if we buy something on credit, it is purchases and accounts payables so, cash is not involved. But later, cash will be involved when we will have to make payment to the creditor so in that case the accounts payables and cash will be there.

So, for various transactions we need to have cash. So, we can say that first motive of holding cash is the Transactional Motive.

Speculative Motive:

At times, we get opportunities and to avail these opportunities (opportunity in terms of business, in terms of making more profit, in terms of getting more return) we need to have sufficient amount of cash or at least have the availability of arranging cash so that we can make effective use of those opportunities.

So, many times we maintain cash for such Speculative purposes that if there are some contingencies then we can utilize them effectively for the operations of our business.

Precautionary measure or Motive:

Precautionary measure means that it happens that whatever we have planned, whatever is our budget, whatever was the anticipated requirement of our cash, may not be our actual requirement; we might have thought about having let's suppose \$1000 cash will be enough for a month but actually we will require more than that because of some emergencies or for some unforeseen things or lack of planning.

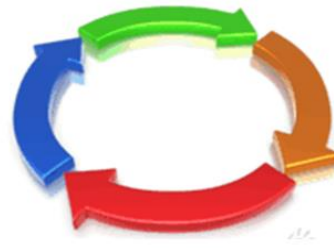
So, it happens that at times we are short of cash because of all these possibilities. If there is something wrong in our planning or calculations or any situation created which was not there when we have started the planning of our cash amount and therefore we keep some amount for safety or Precautionary measures.

Hence, these are the **three common reasons or motives of holding cash** that is for **Transactions**, that **is for utilizing any opportunity** and that **is for keeping ourselves safe** if there are some emergencies.

When we talk about **Cash Management** then generally in the normal course of business, Cash is paid and received. It means that we already know that when we purchase goods or do expense then cash outflow happens and when we sell goods then there is a cash inflow. So, this is a cycle that we are paying cash and we are receiving cash.

So, more we are paying for purchases of our goods and expenses, more will be the outflow of cash and if we are receiving more from our receivable accounts or from sales then more will be our cash inflow.

Cash Cycle



So, there is a cash cycle that cash is coming into revenue & sales and it is going out for expenses and for the purchase of goods in the normal course of business operations. So, we can calculate what is called Cash Conversion Cycle.

This **Cash Conversion Cycle** is basically related to our production or sales cycle in which we have three things meaning that **when we do sales** then there arises our accounts receivables in which our cash get blocked. More will be the credit period of receivables and our cash will be blocked for more time period that is not good for the business organization.

Second thing is that when **we keep inventory** then our cash is also blocked in inventory and longer is the period, more is the quantity of inventory that we are holding then more cash will be involved and for longer period of time.

When we add these two things then we get a figure that **how much amount of cash will be blocked**.

Cash Conversion Cycle - CCC = Days of Sales Outstanding **PLUS Days of Inventory Outstanding**

On the other hand, **we have make the payments** to our suppliers; our creditors from which we have purchased things on credit so, their payments that we have to make, if that amount is large and its credit period increases then we are left with some cash if we have not made payment to them. So, to calculate that Cash Conversion Cycle whose formula is given below:

Cash Conversion Cycle - CCC = Days of Sales Outstanding **PLUS Days of Inventory Outstanding **MINUS** Days of Payables Outstanding**

How we can improve this Cash Conversion Cycle?

Accounts Receivables and Inventories are the two accounts where our cash is blocked and to block cash payables, cash is left with us. So, if we can reduce the credit period or if we can reduce the days for which accounts receivables are there in our business and if we can reduce the holding period of inventory and we can reduce the quantity of inventory then through which our cash cycle will be improved. And similarly, if we increase the period of payment and this will also improve our cash cycle.

To improve the cash cycle, we have to reduce the receivables and inventory amount and we have to increase the payables amounts.

We can conclude that the presence of cash, their uses are of three types; Transactional, Speculative and Precautionary motives and then we have a cash cycle that is created in normal business which is related to our purchases, sales and inventory holding and we have to improve this cash cycle in such a way that we have more cash inflows and less cash outflows in order to keep more cash for longer period for all the measures that we have just discussed in this lecture.

Module 8: Working Capital Management

Topic 162: Cash Management-3

We have discussed that Cash is extremely important to run and operate a business. Cash is collected from several sources and cash is paid to different groups of people. We have discussed about what are the purposes of holding cash and how & when the cash is required.

Now we will talk about very important tool of managing Cash that is called Cash Budget.

Cash Budget:

Budget is something which every person would do budgeting in his/her daily life. It means that everyone has cash and we keep budget of it that how much cash we

will receive and where we have to spend it and how much amount will be left with us and how much cash will be deficit. This planning is done by almost every family and every earning hand. And simply this is called a Budget.

If we are making a plan of managing cash in a way that we want to see that from where we will be collecting cash from our business which is called **sources of cash** and where we will invest that cash or where we will be requiring that cash for payment and then we compare that what we had earlier and what we will have and where we will be needing it and at the end whether we will end up with the surplus or deficit which is simply called **Cash Budget**.

Cash Budget

- Opening Balance
- Receipts
- Payments
- Closing Balance
- Surplus or Deficit



Opening Balance: To make a Cash Budget, first we need to see that what is the Opening or Starting Balance. Suppose we want to make a Cash Budget for the year 2021 or any year and in that year the amount of cash at the beginning of that year is the Opening Balance.

This can be a positive balance or a negative balance. It means that it may be that at the start; we have negative balance or have an overdraft or have an amount which we have to pay that is negative amount.

Receipts: In this Opening Balance, we add the anticipated Receipts. The Receipts which we think will be there meaning that how much cash will come through our sales or our receivables or any other sources so, this is the **Receipts of Cash**.

Payments: We subtract the Payments from this sum of Opening Balance and Receipts meaning that where we have to pay how much amount of cash. So, this will be an outflow of cash and we call it an application of cash.

Closing Balance: And as a result, we will have a Closing Balance. It means that from Opening Balance to Closing Balance, in simplest possible words, we have Payments and Receipts where Receipts increases our balance and Payments decrease our balance of cash.

Surplus or Deficit: And this Closing Balance that we will have at the end of a certain period of time (i.e., one year) this balance may be a surplus or deficit.

Surplus means that we are left with cash meaning that the cash we have planned but we have more cash than planned cash.

And **Deficit** means that there is cash short fall means that the cash we have planned but we have less cash than planned cash. Because it is a budget and these are not actual figures.

So, in advance we would know that whether at the end of a year or if we are making our monthly budget then at the end of the certain month, what will be the balance of our cash. It means that after one month whether we have surplus (left with cash) or deficit (short fall).

Advantages of making Cash Budget:

The advantage of making Cash Budget is that in advance we would know that whether we will be requiring borrowing some amount of cash or we will be no need to borrow but we will have surplus cash which we can plan to invest somewhere to make additional profits.

Sources of Cash:

Sources of cash mean that from where cash or receipts will come.

Sources of Cash

- Sales/Revenue
- Loans/Debts
- Sale of Assets

Sales/Revenue: The main source of cash is Sales or Revenue meaning that if we are selling some product then it is sale and if we are selling some services then it is our revenue. So, this Revenue or Sales whether it is a cash sale or a credit sale but ultimately it results into source of cash.

Loans/Debts: When we take some loan then there is a cash inflow and that is a source of cash.

Sale of Asset: Many times we have some such assets which we don't need them or their value (i.e., land) has increased today than the day of purchase. And now today we can expand our business by selling this fixed asset or land in our business and we will get cash inflow which is a big source of cash.

Applications of Cash:

In contrast to sources, are the applications of cash meaning that where we have to utilize that cash or where we will need that cash and where will be the cash outflows.

Applications of Cash

- Purchases
- Expenses
- Repayment of Loan
- Investment



Purchases: A large amount of cash goes for Purchases the goods, the inventory that we purchase and we have to pay cash for that. So, the major application of cash is in Purchases.

Expenses: We have to incur Expenses either in the production (i.e., factory expenses, labor, salaries, wages, utility expenses, the overheads, marketing expenses, sales expenses, administration expenses etc.) all these expenses require cash. So, a very large sum of cash actually goes for the payment of these Expenses and it is a huge Application of Cash.

Repayment of Loan: We have said that we do cash inflow through borrowing and whenever a business borrows there has to be a return of that amount, there has to be a Repayment of that amount and that Repayment of certain Loan or a Debt or a liability is actually requires a cash outflow.

So, it means that we have Purchases, Expenses and Repayments of loan in Applications of Cash.

Investment: In addition to these, we have another thing which we can call opportunity of making our cash more profitable which is **an Investment Opportunity.** It means that we have a situation that if we have cash then we can earn further profit from that cash and this is Investment Opportunity and Speculative purposes as we have talked about earlier.

So, cash would also be required for investing purposes and that is also an Application of Cash.

These Sources and Applications of Cash actually to the results whether we have more sources or more applications and that means whether we will have Surplus or we will have a Deficit.



It means that if we have more cash inflow or our sources of cash will be more then we will have **Cash Surplus**. If there is Cash Surplus, we can invest that cash somewhere and can make additional profit.

If we have cash outflows will be more and sources or inflows of cash will be less then in that case we have **Cash Deficit**. It means that we have anticipated short fall that after one month let's say we will have shortage of 2 million rupees.

So, we at the beginning of this month, we will start working on arranging that amount of cash from some bank or some other facility so, that we can cover up that Deficit.

It means that this Surplus and Deficit are nothing but the difference of our Sources and Applications of cash. If we have more sources or cash inflows and applications or cash outflows are less then we will have Surplus cash which we will invest somewhere.

If we have more applications or outflows of cash and less sources or inflows of cash then we will have Cash Deficit. And if there will be Cash Deficit then it means that we have to make arrangement of cash in advance to cover that Deficit in order to avoid Technical Insolvency.

So, we can conclude our discussion of this lecture that Cash is needed for various purposes and considering all those purposes, we make a Cash Budget and Cash

Budget is actually about our Sources of Cash and Applications of Cash meaning that from where cash inflows are coming and to where cash outflows will go and where we have how much amount of cash and what amount of cash is left at the start or end. This is what we do in routine regularly.

So, that we in advance are in a very good position to make decisions about our future.

Cash Management is extremely important that any business cannot be run or operated or cannot enhance its profits without the Cash Management and that means the business would not be competitive.

So, certainly Cash Management is as important as Management of Inventory as important as Management of Receivables or any other part of Working Capital.